From Curiosity to Creation: Build Your First AI Project & Research Portfolio

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DAY 1 - What is Intelligence? + How to Do Research

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Camp Objectives: From Zero to One

What You Will Achieve This Week

- **Theoretical Foundation:** Understand the mathematics underlying modern Deep Learning and LLMs.
- **Mechanistic Interpretability:** Stop treating AI as a "black box." We will visualize the exact computational graphs.
- **Original Research:** Formulate a novel question, test it, and document the results.

The Final Deliverable

By Friday, every student will have a fully functional, portfolio-ready AI system (written in Python/PyTorch) accompanied by a formal academic write-up suitable for college admissions or STEM competitions (ISEF, Regeneron).

What is Intelligence? A Formal View

Intelligence in cognitive science and AI is difficult to define uniformly, but generally encompasses:

1. **Perception:** Acquiring real-time data from an environment.
2. **Reasoning:** Deriving logical conclusions from incomplete premises.
3. **Learning:** Updating internal models based on error computation.
4. **Planning:** Optimizing a sequence of actions to maximize a future reward function.

Russell & Norvig's 4 Approaches to AI

- Systems that *think like humans* (Cognitive modeling)
- Systems that *act like humans* (The Turing Test)
- Systems that *think rationally* (Formal logic & syllogisms)
- Systems that *act rationally* (Intelligent agents maximizing utility)

Measuring Intelligence: The Turing Test (1950)

The Imitation Game

Alan Turing proposed a pragmatist approach: If a machine can converse with a human evaluator via a text interface and convince the evaluator that it is human, it exhibits intelligence.

Required Sub-systems for Success:

- Natural Language Processing (NLP)
- Knowledge Representation
- Automated Reasoning
- Machine Learning (to adapt to the evaluator)

Limitations (The Chinese Room)

Philosopher John Searle argued against the Turing Test. If a person inside a room perfectly executes a rulebook to translate Chinese symbols without understanding the language, is that true intelligence? **Syntax is not Semantics.**

The Evolution of AI: Symbolic (GOFAI) vs. Connectionist

Symbolic AI (1950s - 1980s)
"*Good Old-Fashioned AI*" relied on hard-coded logic, rules, and "Expert Systems."

- **Mechanism:** If/Then trees and formal logic (LISP, Prolog).
- **Successes:** IBM's Deep Blue beating Kasparov in Chess (1997). Chess has rigid, mathematically calculable rules.
- **Failures:** It completely failed at tasks humans find trivial, like recognizing a face or understanding a joke, because reality is messy and lacks rigid rules.

Connectionist AI (Deep Learning)
Instead of programming the rules, we program the architecture of a brain, provide massive amounts of data, and force the system to discover its own rules.

The Paradigm Shift

Classical Programming: $Rules + Data = Answers$
Machine Learning: $Answers + Data = Rules$

Moravec's Paradox: The Sensorimotor Reversal

"Hard things are easy; easy things are hard."

High-Level Reasoning (Easy for AI)

- Proving algebraic theorems.
- Statistical forecasting.
- Memorizing the entire internet.

Why? These require massive computational power and memory, which silicon microchips excel at.

Sensorimotor Skills (Hard for AI)

- Walking up a flight of stairs.
- Folding laundry without tearing it.
- Distinguishing a dog from a blueberry muffin.

Why? These skills are the result of millions of years of biological evolution. Humans take them for granted.

Human vs. Machine Learning Mechanisms

Attribute	Human Intelligence	Machine Intelligence (Deep Learning)
Data Efficiency	**Few-Shot Learning:** A toddler needs to see 2 cats to learn the concept.	**Data Hungry:** Needs hundreds of thousands of images to map statistical boundaries.
Knowledge Representation	Abstract, semantic, and contextual. We build physical models of the world.	Dense, continuous vector spaces. Matrices of floating-point numbers.
Generalization	High. Can apply knowledge of riding a bike to riding a motorcycle.	Low (Catastrophic Forgetting). AI trained strictly to play chess cannot play checkers.

Biological Inspiration: The Human Neuron

To build connectionist systems, computer scientists modeled algorithms after biological neuroanatomy.

- **Dendrites:** Receive electrochemical signals from neighboring neurons.
- **Soma (Cell Body):** Accumulates the signals. If the total charge passes a certain threshold, the neuron activates.
- **Axon:** Transmits the output signal to the next network layer.
- **Synapses:** The connections between neurons. Learning occurs via *synaptic plasticity* (connections getting stronger or weaker).

Hebbian Learning

"Neurons that fire together, wire together." We map this biological phenomenon into mathematical space using weight matrices.

The Artificial Neuron (Perceptron)

The foundational unit of an AI model is the **Perceptron**, which performs a linear transformation followed by a non-linear activation.

Linear Combination

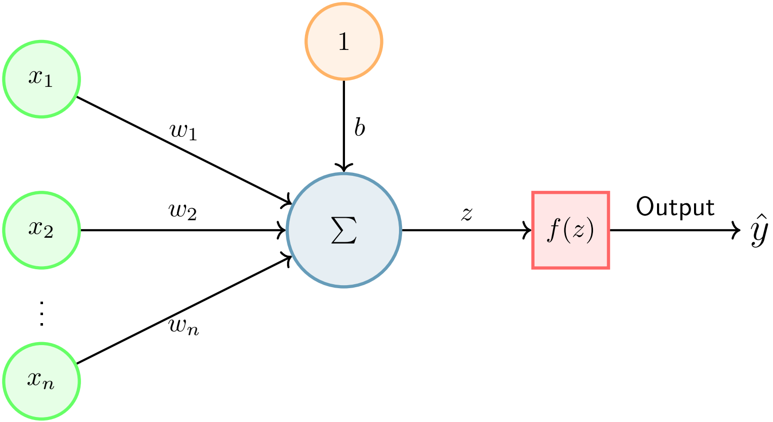
$$z = \sum_{i=1}^{n} w_i x_i + b$$

Matrix Notation:
In modern frameworks like PyTorch, this is computed as a dot product for massive parallelization on GPUs:

$$z = \mathbf{w}^T \mathbf{x} + b$$

- $\mathbf{x} \in \mathbb{R}^n$: **Input Vector** (e.g., pixel intensities).
- $\mathbf{w} \in \mathbb{R}^n$: **Weight Vector** (The learned importance of each input).
- $b \in \mathbb{R}$: **Bias** (Shifts the activation threshold).

Visualizing the Node Computation



Why Do We Need Activation Functions?

The Problem with Linearity

If we do not apply an activation function $f(x)$, a neural network with 100 layers is mathematically useless. It collapses into a single layer.

Proof of Collapse

Let Layer 1 be $y_1 = W_1 x$
Let Layer 2 be $y_2 = W_2 y_1$
Substitute: $y_2 = W_2(W_1 x)$
Since the product of two matrices is just another matrix ($W_{new} = W_2 W_1$):
$y_2 = W_{new} x$

The Solution: Non-Linearity

By wrapping the output in a non-linear function (like σ or ReLU), the network can map complex, curved boundaries in high-dimensional space (e.g., drawing a curve around the pixels that make up a cat's ear).

Core Activation Functions in Modern AI

Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- Squeezes output between 0 and 1.
- Used historically, and currently for binary probability (e.g., "Is this a hotdog? Yes=1, No=0").
- *Flaw:* Vanishing gradients (stops learning on extreme values).

Rectified Linear Unit (ReLU)

$$f(z) = \max(0, z)$$

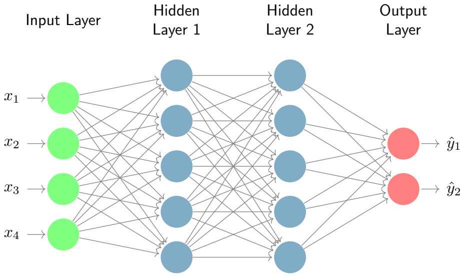
- The industry standard for hidden layers.
- If the input is negative, it outputs 0 (turns the neuron off). If positive, it passes the value through unchanged.
- Highly computationally efficient.

Scaling Up: The Multi-Layer Perceptron (MLP)

To solve complex problems, we stack perceptrons into networks.

- **Input Layer (L_0):** Represents the raw data dimensions.
- **Hidden Layers ($L_1 \dots L_n$):** Extracts hierarchical features. Deep networks have many hidden layers.
- **Output Layer (L_{out}):** Matches the desired prediction format (e.g., 10 nodes to classify the digits 0-9).

A network is "fully connected" if every node attaches to every node in the subsequent layer.



How the Network Learns: The Loss Function

A network learns by making a prediction, measuring how wrong it was, and updating its weights. The measurement of error is the **Loss Function**.

Mean Squared Error (MSE) - For Regression

Used when predicting continuous numbers (e.g., house prices).

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

Where y_i is the true label and $\hat{y}_i$ is the network's prediction.

Cross-Entropy Loss - For Classification

Used when categorizing data into discrete classes (e.g., Cat vs. Dog). Measures the divergence between two probability distributions.

$$\mathcal{L} = -\sum_{i} y_i \log(\hat{y}_i)$$

Optimization: Gradient Descent

Once we calculate the Loss ($\mathcal{L}$), we need to adjust the millions of weights in the network to minimize this error.

- Imagine standing on a foggy mountain and trying to find the lowest valley. You take a step in the direction of the steepest downward slope.
- We compute the partial derivative of the loss with respect to every single weight: $\frac{\partial \mathcal{L}}{\partial w_i}$

The Weight Update Rule

$$w_{new} = w_{old} - \alpha \nabla \mathcal{L}(w_{old})$$

α = **Learning Rate:**

- If α is too high, you overshoot the valley.
- If α is too low, training takes weeks.
- **Backpropagation** is the calculus chain rule algorithm used to compute these gradients efficiently.

The Natural Language Problem: Text to Math

The Translation Barrier
Silicon processors operate on floating-point arithmetic. They cannot parse strings like *"The quick brown fox."* We must map human language into a mathematical space.

Tokenization

The process of breaking raw text into sub-word units (tokens) and mapping them to integer IDs based on a static vocabulary.

Example (GPT-4 Tokenizer):

- "Artificial" $\rightarrow$ [4048]
- " Intelligence" $\rightarrow$ [5154]
- " is" $\rightarrow$ [318]
- " cool!" $\rightarrow$ [3608, 0]

But integers are not enough. The number 5154 does not mathematically relate to 4048. We need geometric representation.

The Naive Approach: One-Hot Encoding

Historically, NLP used **One-Hot Encoding**. If your vocabulary has 50,000 words, every word is represented by a 50,000-dimensional vector of mostly zeros, with a single 1 at the word's index.

$$\vec{v}_{\text{apple}} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^{|V|}$$

Why One-Hot Fails in Deep Learning

1. **Extreme Sparsity:** Wastes massive amounts of RAM.
2. **Curse of Dimensionality:** 50,000 dimensions is too wide for efficient matrix multiplication.
3. **Orthogonality:** The dot product of *any* two different words is 0. $\vec{v}_{\text{dog}} \cdot \vec{v}_{\text{cat}} = 0$. The model has no idea they are related concepts.

The Solution: Dense Vector Embeddings

An **Embedding** compresses a word into a dense, lower-dimensional continuous vector space (e.g., $\mathbb{R}^{768}$).

What do the numbers mean?

The network learns features implicitly. Dimension 1 might represent "royalty," Dimension 2 might represent "gender," Dimension 3 might represent "plurality."

Dense Representation Example:

$$\vec{v}_{\text{king}} = [0.92, -0.45, 0.11, \ldots, 0.88]$$
$$\vec{v}_{\text{apple}} = [-0.12, 0.04, -0.99, \ldots, 0.02]$$

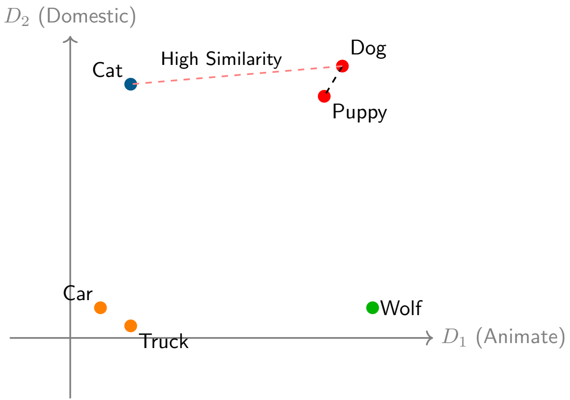
Because the vectors are dense, we can measure semantic similarity using **Cosine Similarity***:*

$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$$

Visualizing Semantic Space ("Vibes in Math Space")

The Distributional Hypothesis:
"You shall know a word by the company it keeps." - J.R. Firth (1957).

- Words with similar meanings cluster together geographically in the embedding space.
- Distance = Semantic similarity.



Vector Arithmetic: Doing Algebra with Concepts

Once language is mapped geometrically, we can perform linear algebra to manipulate semantic meaning.

The Classic Word2Vec Demonstration

$$\vec{v}_{\text{king}} - \vec{v}_{\text{man}} + \vec{v}_{\text{woman}} \approx \vec{v}_{\text{queen}}$$

- Start at the coordinate for **King**.
- Subtract the vector representing **Masculinity**.
- Add the vector representing **Femininity**.
- The nearest geometric neighbor to the resulting coordinate is the word **Queen**.

How LLMs Actually Write: Next-Word Prediction

Large Language Models (like ChatGPT) are formally known as **Auto-Regressive Models**.

- They do not "think" in paragraphs.
- They calculate the probability distribution for the t-th word based on all previous $t-1$ words.

Conditional Probability Formula

$$P(w_t | w_1, w_2, \ldots, w_{t-1})$$

Example:
"The students are sitting in the ____"

- Classroom (88%)
- Library (10%)
- Spaceship (0.01%)

Controlling Creativity: The Softmax Temperature

Before the network outputs a word, it produces raw scores (logits). We convert these into probabilities using the **Softmax function with Temperature (T)**.

$$P_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$$

How T Controls "Vibes":

- $T \to 0$ **(Deterministic):** The model always picks the highest probability word. Highly logical, but repetitive and robotic.
- $T = 1$ **(Default):** Standard distribution.
- $T > 1$ **(Creative/Chaotic):** Flattens the probabilities. The model takes risks and chooses unlikely words, generating poetry—or absolute nonsense.

Demystifying "Research" (From Zero to One)

The High School Myth

"Research means I have to invent a completely new neural network architecture using tensor calculus."

The Academic Reality

Research is systematically answering a *meaningful question* that nobody has perfectly answered yet. It is about observation, isolation, and measurement.

Your Unique Advantage

You do not need to build a 100-Billion parameter model to do AI research.

Models like GPT-4 are massive, unpredictable ecosystems. We treat them like alien organisms. You can do profound research simply by poking the organism and formally recording how it reacts.

When AI Fails: Hallucinations

What is a Hallucination?
When an LLM confidently generates completely false information, fake citations, or nonexistent mathematical proofs. **Why Does It Happen?**

- LLMs are next-word predictors, not databases.
- They have a fundamental drive to *complete the pattern*. If they lack the facts, their probability matrix will invent mathematically plausible (but factually incorrect) tokens to fulfill the prompt.

Research Opportunity

Can you systematically induce a hallucination? Can you design a classifier that detects when the internal probability entropy is high (meaning the model is "guessing")?

When AI Fails: Algorithmic Bias

Garbage In, Garbage Out

- Machine Learning models optimize for the data they are given.
- The training data (the internet) is saturated with human historical prejudice, racial bias, and gender stereotypes.
- The model will mathematically lock in these biases and scale them infinitely.

Real-World Example

In 2018, Amazon scrapped an AI recruiting tool because the model learned to penalize any resume containing the word "women's" (e.g., "Women's Chess Club Captain") because historical engineering hires were predominantly male.

Research Opportunity: How does prompt phrasing alter the gender or racial bias in an LLM's response?

The Scientific Method in AI Research

1. **Observation:** Notice a strange behavior in an AI model.
2. **Hypothesis:** Formulate a testable prediction.
3. **Design the Experiment:** Isolate your variables.
    - **Independent Variable (IV):** What are you changing? (e.g., The emotional tone of the prompt).
    - **Dependent Variable (DV):** What are you measuring? (e.g., The accuracy rate of the math output).
    - **Control Group:** The baseline behavior.
4. **Execute & Log:** Run the prompts, save the outputs.
5. **Analyze:** Use statistics to determine if your change actually caused the result.

Formulating Your Question: The Template

A strong research question is Specific, Testable, and Focused.

The Algorithmic Template

"How does **[Manipulated Variable A]** *affect* **[Measured Output B]** *in* **[Specific AI Architecture]** *?"*

Too Broad / Unscientific ($\times$)

"Will AI become sentient?"

"Is ChatGPT good at coding?"

Rigorous / Testable ($\checkmark$)

"How does few-shot prompting affect the hallucination rate of Llama-3 on calculus derivations?"

How to Read an AI Paper (Without Drowning)

You will need to read literature (arXiv, NeurIPS, ACL) to back up your hypothesis. Do not read them like a novel from start to finish.

The Anatomy of a Paper

1. Abstract
2. Introduction
3. Methodology / Math
4. Results & Tables
5. Conclusion

The Three-Pass Approach:

1. **The Bird's-Eye View (5 mins):** Read only the Title, Abstract, and Conclusion. *What did they do?*
2. **The Mechanics (15 mins):** Look at the architecture diagrams, charts, and tables. *How did they prove it?*
3. **The Deep Dive (1+ hours):** Only do this if the paper is directly related to your project. Read the methodology and the equations carefully.

Day 1 Deliverable & Afternoon Lab

Your Goal for Today (Due at 4:00 PM)

1. **Define Your Research Question:** Use the algorithmic template to draft one original question.
2. **Final Project Concept:** What track are you taking? (Interpretability, Creative AI, or Philosophical Alignment).

Up Next: The PhD-Led Lab Block

We transition from theory to code.

- **Embedding Playground:** Programmatically testing $\vec{v}_{\text{king}} - \vec{v}_{\text{man}} + \vec{v}_{\text{woman}}$.
- **Tiny Neural Network:** Writing the perceptron math in Python.
- **Vibe Classifier:** Training a simple vector-based emotion detector.

Thank you!
Questions?

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